

Say that you want to prove that the language $L = \{M : M \text{ is a TM, } M \text{ has property } P\}$ is undecidable for your favourite property P .

Step 1: Assume not, so there is some decider D that decides L

Step 2: Construct the following decider D_H for HALT

$D_H(\langle M, w \rangle)$:

1. Construct M' where M' runs $M(w)$, & then:

(a) If $M(w)$ halts, the strings M' accepts are s.t M' has property $P \Rightarrow M' \in L$

(b) If $M(w)$ loops, the strings M' accepts are s.t M' does not have $P \Rightarrow M' \notin L$.

Note: You may need

to swap in which case M' has property P , depending on the property P . See the example below!

2. Run $D(M')$:

(a) If $D(M') = 1$, then M' has P , hence $M(w)$ halts. So, ACCEPT.

(b) If $D(M') = 0$, then M' does not have P , hence $M(w)$ loops. So, REJECT.

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Similarly you would then swap here.

(No other case as D is a decider / total TM.)

Step 3: Observe that line 1. in D_{HALT} is always possible for well-defined properties P , & D is total. Hence D_{HALT} is total, & accepts $M(w)$ that halt. Hence D_{HALT} decides HALT. (Contradiction!)

Step 4: HALT has no such decider. Our only assumption was that D exists, hence D cannot exist. Thus, L is undecidable.

You can now plug in any* property P into this framework, & you are done!

(*: "any" is defined in the next page)

Ex: Let $L = E_{TM} = \{M : M \text{ is a TM, } L(M) = \emptyset\}$

Hence, our property P here is that M does not accept any strings.

Thus M' runs $M(w)$ and:

(a) If $M(w)$ halts, M' accepts every string $\Rightarrow L(M') = \Sigma^* \Rightarrow M' \notin L$.

(b) If $M(w)$ loops, M' always rejects every string $\Rightarrow L(M') = \emptyset \Rightarrow M' \in L$

Plugging M' into Step 2 above & making sure to swap what we do when $D(M') = 1$, we are done.

Note: Here we swap as property P is s.t if $M(w)$ loops then M' already has P . Alternatively, you could instead decide $\bar{L}$ & then the entire proof needs no modification, as a decider D for $\bar{L}$ gives a decider for L , namely $\bar{D}$.

Exercise: Do this for $L = \{M : M \text{ only halts on even-length strings}\}$.

② What properties P work? (a.k.a Assignment 2 Q3.(b))

Observe that all properties P do not work; if P is the property that " M halts in one step" then clearly this is decidable, just run M for one step. So, what properties P can we prove undecidability for?

Defⁿ: A non-trivial property P is s.t. $\exists$ Turing machines M_1, M_2 s.t. M_1 has property P & M_2 does not have property P , & $\forall M_3, M_4$ s.t. $L(M_3) = L(M_4)$ we have M_3 has P iff M_4 has P .

A non-trivial property, hence just says that if two TMs compute the same thing, they either both have P or both don't have P , & there are atleast some TMs that have P , but not all, so $P \neq \emptyset$ & $P \neq \Sigma^*$.

Rice's Th^m is the core reason for which properties P are undecidable & which do not.

Th^m: [Rice's Theorem] The language $L = \{M : M \text{ has property } P, M \text{ is a TM}\}$ is undecidable, for all non-trivial properties P .

The proof is a bit long, but not too hard to show, you can look it up if interested.

Don't use this as the answer to the assignment qⁿ! Reason out why non-trivial properties are undecidable.